

Maayan Matsliah

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Education

Northeastern University

Boston, MA

B.S. in Computer Science, AI Concentration | GPA: 3.88/4.0 | Dean's List

Expected May 2029

- **Relevant Coursework:** Mathematical Foundations of AI, Program Design & Implementation II, Discrete Structures, Linear Algebra, Statistics
- **Certificates:** Machine Learning Foundations (Cornell), AWS Cloud Practitioner Essentials (AWS)

Technical Skills

Languages: Python, Java, JavaScript, HTML, CSS, SQL, Shell

Frameworks/Tools: TensorFlow, Keras, PyTorch, torchvision, scikit-learn, Flask, NumPy, Pandas, Matplotlib, Seaborn, DuckDB, REST APIs

ML/AI Concepts: RAG, Agentic AI Workflows, Convolutional Neural Networks, Supervised & Unsupervised Learning, LangChain, Data Encoding, Trees, KNN, Feature Engineering, Model Evaluation, Computer Vision, ResNet, Word2Vec, Hyperparameter Tuning

Experience

Cornell Tech

Break Through Tech AI/ML Fellow

Mar 2026 – Present

- Trained and benchmarked Decision Tree and neural network classifiers on Airbnb price prediction (81.0% vs. 82.0% accuracy) and chose the Decision Tree for production, cutting training time ~100x with no meaningful accuracy loss.
- Built a RAG-powered chatbot using LangChain, Word2Vec, and OpenAI's API for a camping company, grounding responses in their internal files so the model can answer business-specific questions beyond its base knowledge.
- Matched to a 6-person AI Studio team with MathWorks, building a fault-diagnosis model for smart appliances, applying pruning and quantization to compress it for embedded deployment.

GitHub

Open Source Contributor

Jan 2026 – Present

- Authored and merged 12 pull requests across production codebases with 200–700+ GitHub stars, including `conda/constructor`, `python-humanize/humanize`, and the Munich Quantum Toolkit.
- Refactored a scientific-notation utility in `python-humanize`, simplifying the logic and improving benchmarked performance by 5% (75.3µs → 71.5µs).
- Identified and fixed a silent error-handling bug in a production data platform by adding proper API response unwrapping, ensuring failed requests surface accurate errors instead of false success messages — reviewed and approved by 5 maintainers before merge.

Princeton Neuroscience Institute (FlyWire)

Citizen Scientist & Internship Coordinator

May 2023 – Present

- Designed and ran a weekly progress-report system across 15 interns, analyzing self-reported status to flag at-risk progress and surface common blockers; co-led resulting 1:1 check-ins with a Princeton researcher, contributing to 11 of 15 interns being cleared to present to 30 scientists at the Princeton Neuroscience Institute symposium.
- Designed weekly sync-up meeting structure and organized talking points based on common questions and roadblocks.
- Annotated neuron type and location for the BANC connectomics dataset using the Codex AI platform.

Technical Projects

Citation Graph Analysis | *Python, SQL, DuckDB, OpenAlex API, Claude Code* | GitHub

- Independent research using a citation graph to test how field diversity and the rise of remote work reshaped scientific collaboration.
- Architected a high-throughput graph processing pipeline in Python and DuckDB to ingest 413M+ JSON records and process a 2.9B-edge citation graph under strict memory constraints (12GB RAM).
- Designed partitioned batching algorithms to detect reciprocal edges across ~2.9B unnested citations, reducing memory footprint by >90% while achieving 100% referential integrity validation.
- Engineered automated data-cleaning and taxonomy backfill pipelines resolving left-boundary metadata censorship across 34M historical works and enabling rapid decile aggregation across 115M+ papers.

Marine Debris Classifier | *Python, PyTorch, torchvision, ResNet18* | GitHub

- A computer vision classifier built to support automated ocean cleanup by distinguishing marine life from debris.
- Engineered an end-to-end data pipeline in PyTorch across 3 public and custom datasets (~26K underwater images), resolving an initial 10:1 class imbalance down to 3:1 and applying dynamic augmentations (random resized cropping, horizontal flips, ImageNet normalization) to prevent overfitting.
- Fine-tuned a pretrained ResNet-18 classifier using SGD with momentum (0.9) and StepLR decay, overcoming an initial 46% baseline to achieve 99.81% validation accuracy across 10 epochs on an NVIDIA T4 GPU.
- Evaluated class-level performance via Scikit-Learn confusion matrices and classification reports, prioritizing high marine-life recall to minimize false debris classifications and prevent wildlife capture in autonomous ocean cleanup systems.

Savings Made Simple | *Python, Flask, HTML, Render, SQLAlchemy* | GitHub | Live Demo

- A personal budgeting tool built to track spending against a savings goal while studying abroad and guide real-time budget adjustments.
- Built and deployed a Flask/SQLAlchemy web app on Render with relational data models, session-managed CRUD, and dynamic weekly burn-rate calculations.
- Engineered in-memory Matplotlib visualizations (Base64/BytesIO, no disk I/O) and pytest suites covering financial projection edge cases.